

T) Funcion de transferencia

Modelando a partir de la consecuencia de fuerzas

$$F_r(x) + F_f(x) + F_l(x) = F_e(x)$$

donde:

$$F_r(x) = kx(x)$$

$$F_f(x) = c \frac{dy(x)}{dt}$$

$$F_l = m \frac{d^2 y(x)}{dt^2}$$

Entonces

$$m \frac{d^2 y(x)}{dt^2} + c \frac{dy(x)}{dt} + kx(x) = F_e(x) = x(x)$$

Usando la transformada de Laplace tomamos que:

$$\mathcal{L} \left\{ \frac{d^2 y(x)}{dt^2} \right\} + \mathcal{L} \left\{ \frac{dy(x)}{dt} \right\} + \mathcal{L} \{ kx(x) \} = \mathcal{L} \{ x(x) \}$$

$$ms^2 y(s) + c s y(s) + k y(s) = x(s)$$

Funcion de transferencia
mas resorte

Funcion de transferencia circuito

Unica malla

$$-V_i(x) + L \frac{d i_1(x)}{dt} + \frac{1}{C} \int_0^x (i_L(x) - i_2(x)) dt = 0$$



COLEGIOS
MINUTO DE DIOS

y la place tenemos

$$V_i(s) = Ls I_1(s) + (I_1(s) - I_2(s)) \frac{1}{Cs} \quad *$$

para la malla 2

$$I_2(s)R + \frac{1}{C} \int_0^s (i_2(x) - i_1(x)) dx = 0 ; V_o(s) = i_2(s)R$$

a la place

$$I_2(s)R + (I_1(s) - I_2(s)) \frac{1}{Cs} = 0$$

$I_2(s)$, tenemos:

$$\frac{I_1(s)}{Cs} = \frac{I_2(s)}{Cs} + I_2(s)R$$

$$I_1(s) = \frac{I_2(s)}{Cs} + I_2(s)R$$

$$I_1(s) = I_2(s) (1 + CRs) \quad **$$

Ahora (**) en (*): tenemos

$$V_i(s) = Ls I_2(s) (1 + CRs) + (I_2(s) (1 + CRs) - I_2(s)) \frac{1}{Cs}$$

$$V_i(s) = Ls I_2(s) + CR Ls^2 I_2(s) + (I_2(s) + CRs I_2(s) - I_2(s)) \frac{1}{Cs}$$

$$V_i(s) = Ls I_2(s) + CR Ls^2 I_2(s) + R I_2(s)$$

$$V_i(s) = I_2(s) [CR Ls^2 + Ls + R]$$

$$\frac{I_2(s)}{V_i(s)} = \frac{R}{CR Ls^2 + Ls + R}$$

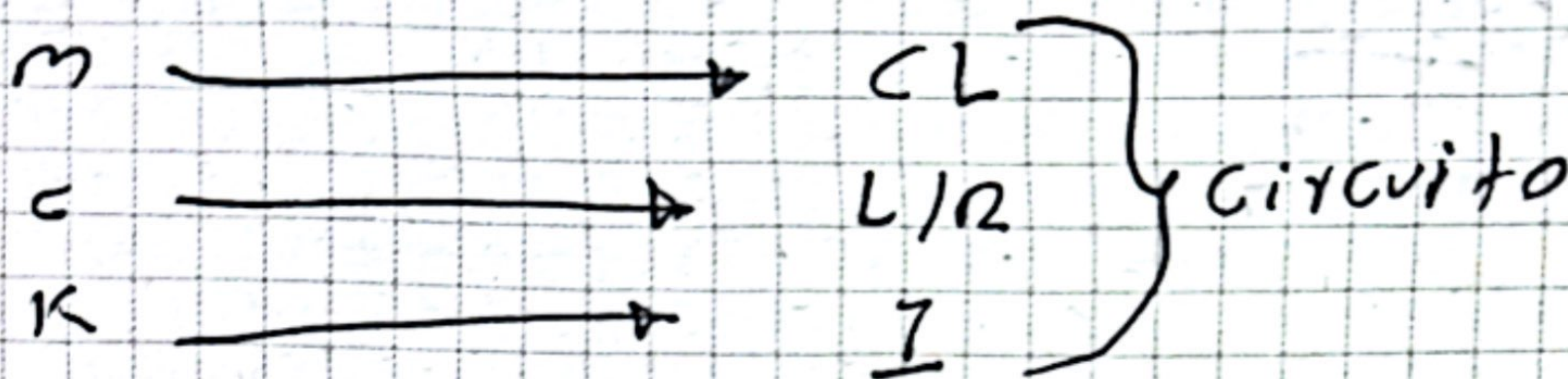
$$\frac{V_o(s)}{V_i(s)} = \frac{R}{CR Ls^2 + Ls + R} \cdot \left(\frac{1/R}{1/R} \right)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{CLs^2 + Ls + R}$$

funcion de transferencia
del circuito electrico.

el circuito electrico es equivalente al sistema
de masa resorte.

Sistema de masa - resorte



por lo tanto

$$H(s) = \frac{1}{Lcs^2 + L/Rs + 1}$$

el equivalente es

$$H(s) = \frac{1}{ms^2 + cs + k} = \frac{1/m}{s^2 + cs/m + k/m}$$

Buscando la forma canonica de segundo orden:

Comparando terminos

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + cs/m + k/m$$

$1=1$ Coeficientes de s^2

$2\zeta\omega_n = \frac{c}{m}$ Coeficiente de s

$\omega_n^2 = k/m$ coeficiente independiente



• Frecuencia natural no amortiguada

$$\omega_n = \sqrt{K/m}$$

• factor de amortiguamiento

$$2 \xi \sqrt{K/m} = c/m$$

$$\xi = \frac{c}{2m\sqrt{K/m}}$$

• Ganancia K

$$K \omega_n^2 = 1/m$$

$$K = \frac{1}{m \omega_n^2}$$

$$K = \frac{1}{m \frac{K}{m}}$$

$$K = \frac{1}{K}$$

La forma canónica de segundo orden:

$$H(s) = K \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$H(s) = \frac{1}{K} \frac{K/m}{s^2 + 2\left(\frac{c}{2m\sqrt{K/m}}\right)\sqrt{K/m} + K/m}$$

$$H(s) = \frac{1/m}{s^2 + c/m s + K/m}$$

$$H(s) = \frac{1}{m(s^2 + c/m s + K/m)}$$

